Estimating a Rational Inattention Discrete Choice Model: A Semiparametric Approach

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Background

- Individuals often choose among alternatives without fully knowing their payoffs.
- Assessing those payoffs requires cognitive effort.
- The theory of rational inattention (RI), pioneered by Sims (2003), models how individuals choose what to learn about uncertain payoffs before deciding.
- The theory of rational inattention is about optimal information acquisition.

Background: an empirical example

- How does labor supply elasticity change when information becomes easier to acquire?
- I illustrate my method using the data of Dubé et al. (2020) from Amazon Mechanical Turk.
  - How does better information about a job's true difficulty affect how the probability of taking it responds to wages?

Background: an Amazon MTurk posting

“Transcribe this MP3 of a 5 minute interview”

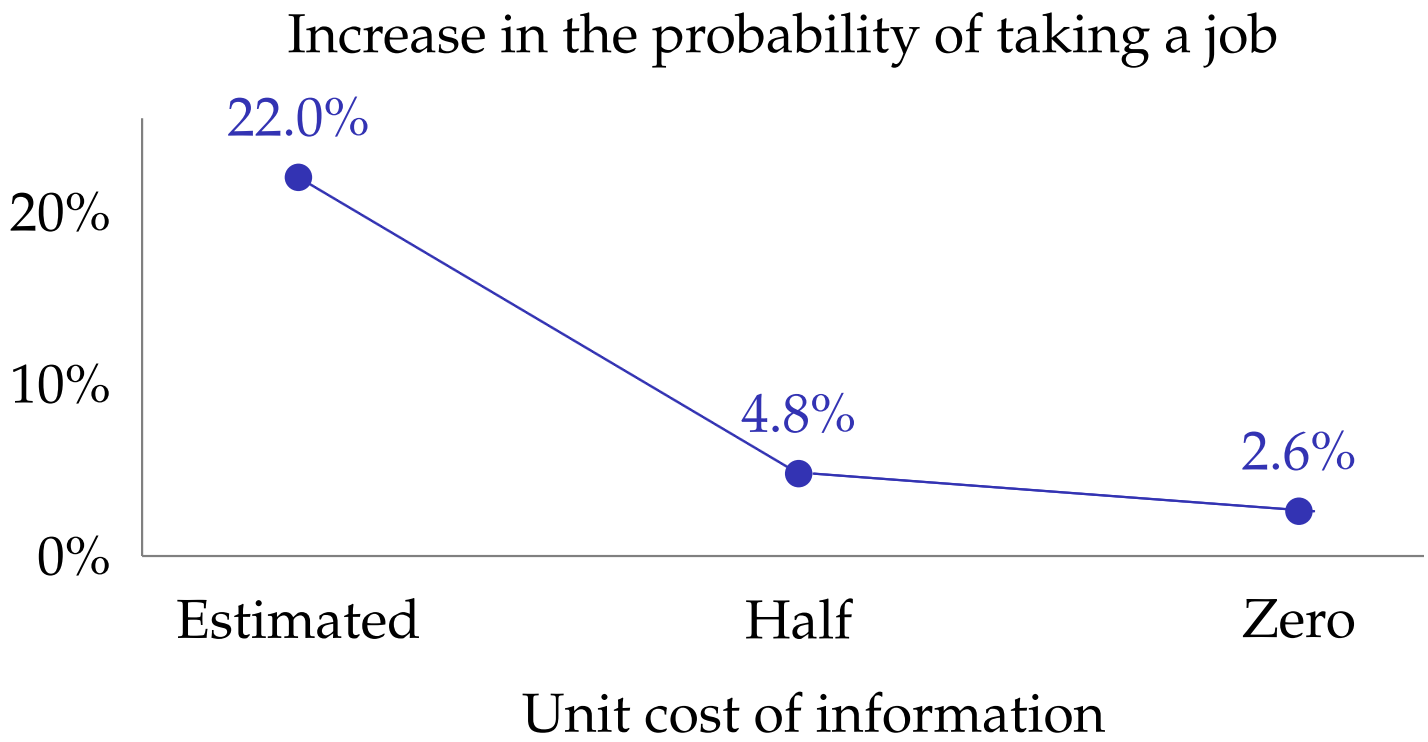
Posted reward: $4.00.

- The same assignment can be manageable for one worker and demanding for another.
- Reading the postings produces an internal assessment of the task’s personal difficulty.

Source: Dubé et al. (2020).

Background: Preview of empirical results

When wages double, how much does the probability of taking a job increase?



Measured using the specificity of job descriptions (a proxy).

Research Question

- Suppose an analyst observes the choices of rationally inattentive individuals who are initially uncertain about their taste shocks, the idiosyncratic payoff components unobserved by the analyst.

- How can exogenous variation in choice data identify the taste-shock distribution and how the unit cost of information varies with observed characteristics, without imposing a parametric form on either?

This paper

- **Main contribution: constructive semiparametric identification**
  - Utility and information-cost shifters identify the payoff coefficients, the taste-shock distribution, and the unit cost of information at each value of the cost shifter.
  - I impose no parametric form on the taste-shock distribution or on how the unit cost of information varies with observed characteristics.
- **Secondary contribution: estimation and inference**
  - I propose a consistent sieve minimum-distance estimator using market shares.
  - I construct confidence regions by test inversion, without requiring asymptotic normality of the estimator.

Related literature

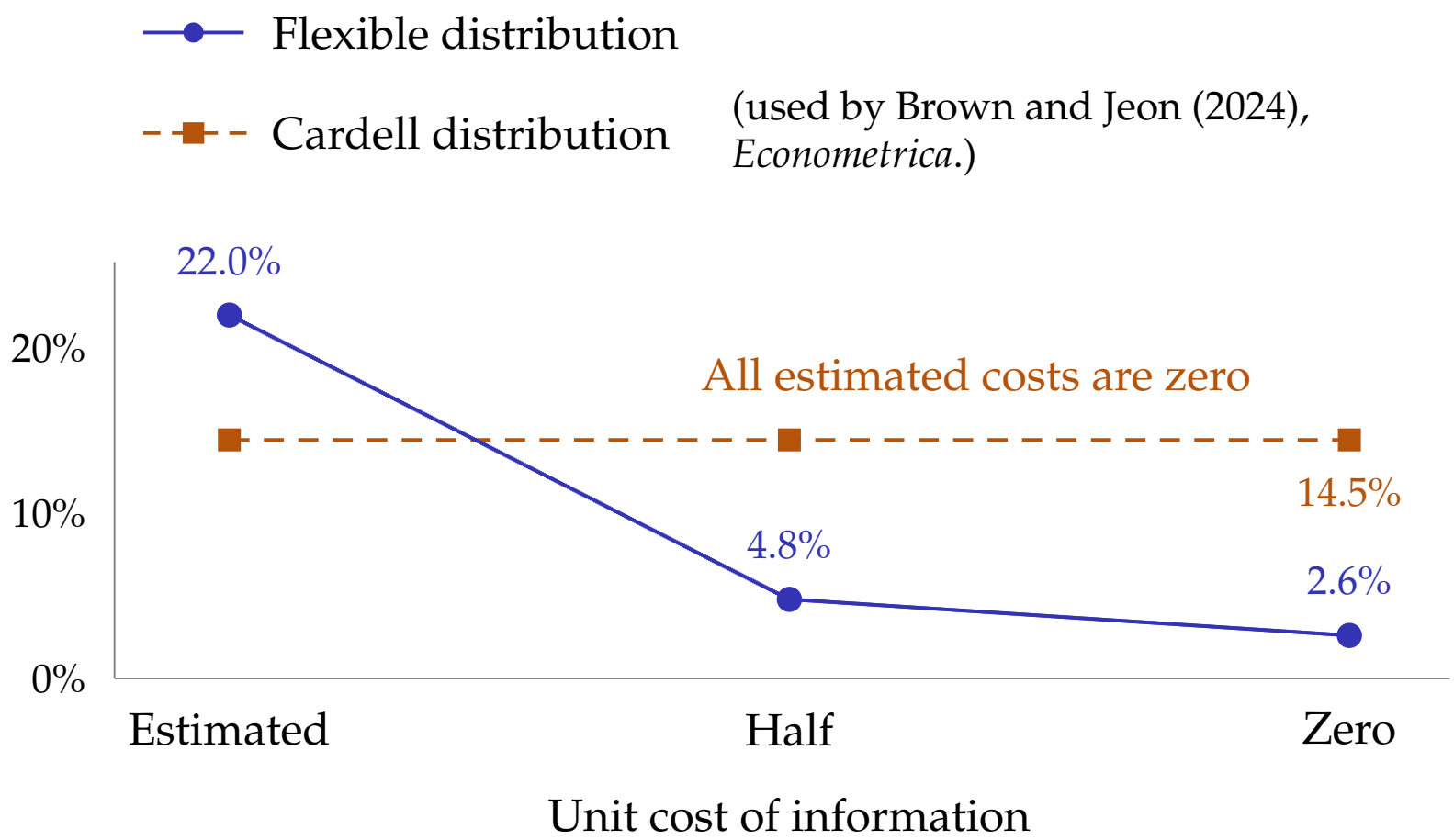
- **RI theory and revealed preference**
  Sims (2003); Matějka and McKay (2015); Caplin and Dean (2015); Caplin, Dean, and Leahy (2022); Fosgerau et al. (2020); Afrouzi (2024); Angeletos and Sastry (2025); Hansen et al. (2025); Maćkowiak and Wiederholt (2025, 2026).
- **Consideration sets**
  Caplin, Dean, and Leahy (2019); Masatlioglu, Nakajima, and Ozbay (2012); Manzini and Mariotti (2014); Cattaneo et al. (2020).
- **Empirical RI**
  Brown and Jeon (2024); Song and Stern (2025); Weber et al. (2025).
- **Econometrics of RI and imperfect information**
  Brown and Jeon (2024); Liao (2024); Gualdani and Sinha (2024); Joo (2023).
- **Semiparametric identification of discrete choice models**
  Lewbel (1997), "Semiparametric Estimation of Location and Other Discrete Choice Moments."

Relation to existing work

- Brown and Jeon (2024) estimate insurance demand with an RI model, using a parametric Cardell prior over costly-to-learn taste shocks and a parametric information-cost function.
- Liao (2024) studies identification with market-level uncertainty in a parametric family; Joo (2023) develops a parametric RI demand model with subjective priors.
- This paper identifies the individual-specific payoff distribution and the information-cost function nonparametrically, and estimates the model using aggregate choice data.

A flexible specification of the taste distribution matters

The two fitted models imply different wage responses when information costs fall.



RI model: the agent's decision problem

- The individual chooses between alternatives $j \in \{0, 1\}$ with random payoffs $U := (U_0, U_1)$ and prior distribution P_U.
- An information structure specifies how likely she is to receive each signal at each possible payoff. She chooses it before receiving the signal.
- Following Matějka and McKay (2015), I represent this structure by $F := F_{T,U}$, the joint distribution of a signal T and payoffs U.
- She chooses F subject to $F_U = P_U$. Write F_T for its signal marginal.
- After receiving $T = t$, she chooses the alternative maximizing

$$\mathbb{E}_F[U_j \mid T = t].$$

RI model: the decision problem (continued)

$$\max_{F:\, F_U = P_U} \left\{ \underbrace{\mathbb{E}_{F_T}\left[\max_j \mathbb{E}_F[U_j \mid T]\right]}_{\text{expected payoff after learning}} - \underbrace{\lambda I_F(U;T)}_{\text{information cost}} \right\}.$$

- Shannon mutual information measures the amount of information the signal conveys about the payoffs:

$$I_F(U;T) := \int \log\left(\frac{dF}{d(F_T \otimes P_U)}\right) dF.$$

- Multiplying this amount by the unit cost of information $\lambda \geq 0$ gives the information cost $\lambda I_F(U;T)$.

RI model: payoffs and information costs

For individual i in market m, observed shifters satisfy $Z_m \in \mathbb{R}$ and $X_m \in \mathbb{R}^{d_x}$, where d_x is a positive integer.

$$U_{i0m} = 0, \qquad U_{i1m} = \underbrace{\alpha + \delta Z_m}_{\text{known before learning}} + \underbrace{\Omega_{im}}_{\text{costly to learn}}.$$

- Write $\mathcal{X} := \text{supp}(X_m)$. Let $\lambda : \mathcal{X} \to [0, \infty)$ give the unit cost of information, with $\lambda_m := \lambda(X_m)$.
- Ω_{im} is i.i.d. across i, m with distribution G. Also, Ω_{im} is independent of (Z_m, X_m).
- Individuals know G, their common prior over the taste shocks, before acquiring information.
- Write $\theta := (\alpha, \delta, \lambda, G)$ for a parameter value.

RI model: choice probabilities

- At $\lambda(x) > 0$, the ex ante probability of choosing alternative 1 under θ, before receiving the signal, is

$$b(z, x; \theta) := \arg\max_{q \in [0,1]} \int \log\left(1 - q + q e^{(\alpha + \delta z + \omega)/\lambda(x)}\right) dG(\omega).$$

  This representation follows from Matějka and McKay (2015).

- A nondegenerate G makes the objective strictly concave in q, so the maximizing probability is unique.

RI model: conditional choice probabilities

- At $\lambda(x) > 0$ and $0 < b(z, x; \theta) < 1$, the probability of choosing alternative 1 conditional on $\Omega = \omega$ is

$$p_\theta(\omega; z, x) := \frac{b(z, x; \theta) e^{(\alpha + \delta z + \omega)/\lambda(x)}}{1 - b(z, x; \theta) + b(z, x; \theta) e^{(\alpha + \delta z + \omega)/\lambda(x)}}.$$

- Averaging over the taste-shock distribution gives the ex ante probability:

$$b(z, x; \theta) = \int p_\theta(\omega; z, x) \, dG(\omega).$$

  For $0 < b(z, x; \theta) < 1$, this equation is equivalent to the first-order condition for the preceding maximization problem.

Identification: definition

- Let $\theta_0 := (\alpha_0, \delta_0, \lambda_0, G_0)$ generate the observations $\{(\widehat{s}_m, Z_m, X_m)\}_{m=1}^{M}$, where

$$\widehat{s}_m := \frac{1}{N_{m,M}} \sum_{i=1}^{N_{m,M}} Y_{im}, \qquad (Z_m, X_m) \overset{\text{i.i.d.}}{\sim} F_{Z,X}.$$

- Conditional on market characteristics, choices are independent across individuals and markets, with

$$Y_{im} \mid (Z_m, X_m) \sim \text{Bernoulli}\big(\pi(Z_m, X_m)\big),$$

where

$$\pi(z, x) := \mathbb{P}(Y = 1 \mid Z = z, X = x) = b(z, x; \theta_0).$$

- Let $\mathcal{Z}_x := \text{supp}(Z \mid X = x)$.

Definition. The parameter θ_0 is identified if a known mapping $\mathcal{T}$ satisfies

$$\theta_0 = \mathcal{T}\left(\left\{\{\pi(z, x)\}_{z \in \mathcal{Z}_x}\right\}_{x \in \mathcal{X}}\right).$$

Identification: Assumptions

Assumption ID1

(a) $\mathbb{E}_{G_0}\Omega = 0$, $\mathrm{Var}_{G_0}(\Omega) = 1$.

(b) $\delta_0 \neq 0$.

(c) $\lambda_0 : \mathcal{X} \to [0, \infty)$ is continuous.

(d) For every x with $\lambda_0(x) = 0$, the function $z \mapsto b(z, x; \theta_0)$ is continuous on $\mathcal{Z}_x$.

Assumption ID2: Arbitrarily Low Information Costs

$$\inf_{x \in \mathcal{X}} \lambda_0(x) = 0.$$

Identification: Support of the Utility Shifter

Assumption ID3

(a) For each x, $Z \mid X = x$ has a continuous distribution with interval support $\mathcal{Z}_x$.

(b) Define $\mathcal{X}_0 := \{x \in \mathcal{X} : \lambda_0(x) = 0\}$. For every $x \in \mathcal{X}_0$,

$$\mathbb{P}\{-\Omega \in \alpha_0 + \delta_0 \operatorname{int}(\mathcal{Z}_x)\} = 1.$$

For every $x \notin \mathcal{X}_0$, there exists

$$z_x^* \in \operatorname{int}(\mathcal{Z}_x), \qquad 0 < \pi(z_x^*, x) < 1.$$

Identification: main result

Theorem 1. Under Assumptions ID1–ID3, the true parameter $\theta_0 = (\alpha_0, \delta_0, \lambda_0, G_0)$ is identified.

Identification: method of Lewbel (1997)

- Consider the binary choice model:

$$Y = \mathbf{1}\{\alpha_0 + \delta_0 Z \geq -\Omega\}, \qquad \Omega \perp Z, \quad \delta_0 > 0.$$

- Equivalently,

$$Y = \mathbf{1}\{S \leq Z\}, \qquad S = -\frac{\alpha_0 + \Omega}{\delta_0}.$$

- By definition of S, $\alpha_0 + \delta_0 S = -\Omega$. We call S the switch point.
- Under the normalizations in ID1,

$$\mathbb{P}(Y = 1 \mid Z = z) = F_S(z), \qquad \mathbb{E}S = -\frac{\alpha_0}{\delta_0}, \quad \text{Var}(S) = \frac{1}{{\delta_0}^2}.$$

Identification: extending Lewbel (1997) to RI

- Write $\pi(z,x) = \mathbb{P}(Y = 1 \mid Z = z, X = x)$. At positive cost and an interior share, the RI choice equation gives

$$\boxed{\pi(z,x) = \mathbb{P}\{h_x(z) \geq -\Omega + \lambda_0(x)\varepsilon\}.}$$

  Here ε is standard logistic, independent of $\Omega \sim G_0$, and

$$h_x(z) = \alpha_0 + \delta_0 z + \lambda_0(x)\ell(\pi(z,x)), \qquad \ell(p) = \log\frac{p}{1-p}.$$

- For the complete choice curve, define

$$I_x := \{z \in \mathbb{R} : 0 < \pi(z,x) < 1\}.$$

  Then $h_x : I_x \to \mathbb{R}$ is strictly increasing and onto.

Identification: the RI switch point

- For positive information cost, define

$$\boxed{S_x := h_x^{-1}(-\Omega + \lambda_0(x)\varepsilon)\,.}$$

- Because h_x is increasing,

$$\begin{aligned}\mathbb{P}(S_x \leq z) &= \mathbb{P}\{-\Omega + \lambda_0(x)\varepsilon \leq h_x(z)\} \\ &= \pi(z, x).\end{aligned}$$

Thus the choice curve is the CDF of the constructed switch point.

Identification proof: the switch point identity

- By construction, $h_x(S_x) = -\Omega + \lambda_0(x)\varepsilon$. Hence

$$\boxed{-\Omega + \lambda_0(x)\varepsilon = \alpha_0 + \delta_0 S_x + \lambda_0(x)\ell(\pi(S_x, x)).}$$

- At positive cost, the CDF of S_x is continuous, so

$$\pi(S_x, x) = F_{S_x}(S_x) \sim \text{Uniform}(0, 1).$$

- $\ell(\pi(S_x, x))$ is standard logistic, with mean zero and variance $\pi^2/3$.

Identification proof: linking moments to parameters

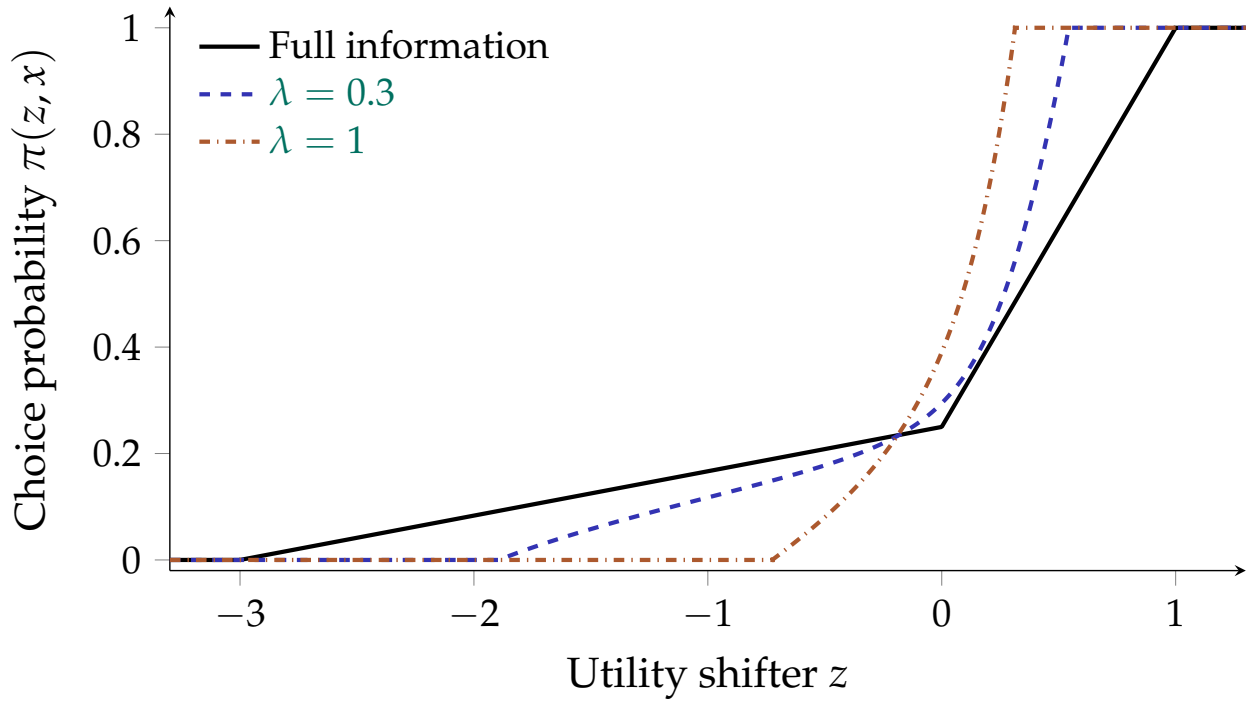
- The choice curve determines the following switch-point moments:

$$\mu(x) = \mathbb{E}S_x, \qquad V(x) = \mathrm{Var}(S_x),$$
$$C(x) = \mathrm{Cov}(S_x, \ell(\pi(S_x, x))) > 0 \quad \text{at positive cost.}$$

- Taking means and variances in the switch-point identity gives

$$\mu(x) = -\frac{\alpha_0}{\delta_0}, \qquad \boxed{\delta_0^2 V(x) + 2\delta_0\lambda_0(x)C(x) = 1.}$$

Identification: illustrating switch-point dispersion



Identification proof: recovering the utility slope

- The switch-point variance reaches its upper bound at zero information cost:

$$V(x) \leq \frac{1}{\delta_0^2}, \qquad V(x) = \frac{1}{\delta_0^2} \iff \lambda_0(x) = 0.$$

- ID2 allows information costs to approach zero, so the largest limiting variance identifies the utility slope:

$$\sup_{x \in \mathcal{X}} V(x) = \frac{1}{\delta_0^2}, \qquad \boxed{\delta_0 = \left(\sup_{x \in \mathcal{X}} V(x) \right)^{-1/2}.}$$

Identification proof: recovering the intercept and costs

- With δ_0 identified, the common mean identifies the intercept:

$$\alpha_0 = -\delta_0 \mu(x).$$

- The variance identity identifies the information cost:

$$\lambda_0(x) = \begin{cases} 0, & V(x) = \delta_0^{-2}, \\ \dfrac{1 - \delta_0^2 V(x)}{2\delta_0 C(x)}, & V(x) < \delta_0^{-2}. \end{cases}$$

Identification proof: recovering the taste distribution

- At an x with positive cost, h_x and the distribution of S_x are identified. Thus we know the distribution of

$$h_x(S_x) = -\Omega + \lambda_0(x)\varepsilon.$$

- Independence identifies the characteristic function of $-\Omega$:

$$\boxed{\varphi_{-\Omega}(t) = \frac{\varphi_{h_x(S_x)}(t)}{\varphi_\varepsilon(\lambda_0(x)t)}.}$$

Identification proof (multiple alternatives): reducing to binary comparisons

$$v_0 = 0, \qquad v_j = \alpha_j + \delta_j z_j + \Omega_j, \quad j = 1, \ldots, J.$$

- Information cost $\lambda(x)$ is common; tastes have joint distribution G.
- With only the default and option j chosen, the problem reduces to

$$\max_{p \in [0,1]} \lambda(x) \mathbb{E} \log\left(1 - p + p e^{v_j / \lambda(x)}\right).$$

  - This is the binary model for marginal taste Ω_j.
  - Low enough rival utilities eliminate rival shares under the tail conditions.
- At positive cost, unique extension recovers these binary curves from an observed open region with all shares positive.

Identification proof (multiple alternatives): recovering payoff coefficients

Normalize $\mathrm{Var}(\Omega_1) = 1$.

- The binary comparison between option 1 and the default identifies α_1, δ_1, and the common $\lambda(x)$.
- For another margin, let $\sigma_j^2 = \mathrm{Var}(\Omega_j)$. Then

$$\delta_j^2 V_j(x) + 2\delta_j \lambda(x) C_j(x) = \sigma_j^2.$$

Subtract this equation at two environments with different variances:

$$\delta_j = \frac{2\{\lambda(x')C_j(x') - \lambda(x)C_j(x)\}}{V_j(x) - V_j(x')}, \qquad \alpha_j = -\delta_j \mu_j(x).$$

Identification proof (multiple alternatives): choice probabilities reveal a joint CDF

- At a positive-cost x where all shares are positive, the conditional logit has a representation using independent standard Gumbel random variables $\eta_0, \ldots, \eta_J$.

$$\mathbb{P}(Y = 0 \mid z, x) = \mathbb{P}\left\{\Omega_j + \lambda(x)(\eta_j - \eta_0) \leq c_j(z, x)\ \forall j\right\},$$

where the identified thresholds are

$$c_j(z, x) = -\alpha_j - \delta_j z_j + \lambda(x) \log \frac{b^0(z, x)}{b^j(z, x)}.$$

Writing $\zeta_j = \eta_j - \eta_0$ and $H_x = \Omega + \lambda(x)\zeta$,

$$b^0(z, x; \theta_0) = F_{H_x}(c(z, x)).$$

Identification proof (multiple alternatives): recovering the joint taste distribution

- An observed open region with all shares positive gives an open set of thresholds $c(z, x)$.
- F_{H_x} on that open set identifies its analytic density everywhere.
- Independence gives $\varphi_{H_x}(t) = \varphi_G(t)\varphi_\zeta(\lambda(x)t)$.

$$\varphi_\zeta(t) = \Gamma\left(1 + i\sum_{j=1}^{J} t_j\right)\prod_{j=1}^{J}\Gamma(1 - it_j) \neq 0.$$

$$\varphi_G(t) = \frac{\varphi_{H_x}(t)}{\varphi_\zeta(\lambda(x)t)}.$$

Estimation: the RI share map

Write $W_m = (Z_m, X_m)$, $w = (z, x)$, and $s_m = \pi(W_m)$.

The true parameter satisfies

$$\boxed{r(\theta_0; W_m, s_m) = 0,}$$

where, for an input share q,

$$r(\theta; w, q) = \Psi(\theta; w, q) - q,$$

and

$$\Psi(\theta; w, q) = \begin{cases} \displaystyle\int \frac{q e^{(\alpha+\delta z+\omega)/\lambda(x)}}{1 - q + q e^{(\alpha+\delta z+\omega)/\lambda(x)}}\, dG(\omega), & \lambda(x) > 0, \\ \mathbb{P}_{\Omega\sim G}(\alpha + \delta z + \Omega > 0), & \lambda(x) = 0. \end{cases}$$

Estimation: estimator

- Replace population shares with sample shares in the model restriction:

$$\widehat{Q}_M(\theta) = \frac{1}{M} \sum_{m=1}^{M} r(\theta; W_m, \widehat{s}_m)^2.$$

- Over a finite-dimensional sieve $\Theta_M \subseteq \Theta$, choose an approximate minimizer:

$$\widehat{Q}_M(\widehat{\theta}_M) \leq \inf_{\theta \in \Theta_M} \widehat{Q}_M(\theta) + o_p(1).$$

Estimation: assumptions

- Each market share averages $N_{m,M}$ individual choices:

$$\widehat{s}_m = \frac{1}{N_{m,M}} \sum_{i=1}^{N_{m,M}} Y_{im}.$$

Assumption E1. The market sizes satisfy

$$M \to \infty, \qquad N_M := \min_{m \leq M} N_{m,M} \to \infty.$$

Estimation: Parameter Space and Regularity

Assumption E2

(a) Θ contains θ_0 and is compact under

$$d(\theta, \widetilde{\theta}) := |\alpha - \widetilde{\alpha}| + |\delta - \widetilde{\delta}| + \|\lambda - \widetilde{\lambda}\|_\infty + d_L(G, \widetilde{G}),$$

where d_L is the Lévy metric for weak convergence.

(b) Every $\theta = (\alpha, \delta, \lambda, G) \in \Theta$ satisfies

$$\mathbb{E}_G \Omega = 0, \qquad \mathrm{Var}_G(\Omega) = 1, \qquad \delta \neq 0.$$

The function $\lambda : \mathcal{X} \to [0, \infty)$ is continuous, with

$$\inf_{x \in \mathcal{X}} \lambda(x) = 0.$$

Estimation: Zero Information Cost and Approximation

Assumption E2 (continued)

(c) For every $\theta \in \Theta$ and every $x \in \mathcal{X}$ with $\lambda(x) = 0$, either

- $\mathcal{Z}_x = \mathbb{R}$; or
- $z \mapsto b(z, x; \theta)$ is real analytic on $\mathbb{R}$.

Assumption E3: Sieve Approximation

A deterministic sequence $\{\theta_M^*\}_{M=1}^{\infty}$ satisfies

$$\theta_M^* \in \Theta_M \subseteq \Theta, \qquad d(\theta_M^*, \theta_0) \to 0 \quad (M \to \infty).$$

Estimation: consistency theorem

Theorem 2. Under ID1–ID3 and E1–E3,

$$d(\widehat{\theta}_M, \theta_0) \to_p 0 \qquad \text{as } M \to \infty.$$

Estimation proof: replacing estimated shares

- Let Q_M be the objective evaluated at population shares:

$$Q_M(\theta) = \frac{1}{M} \sum_{m=1}^{M} r(\theta; W_m, s_m)^2.$$

- Under E1, using estimated shares changes the objective by a negligible amount, uniformly over candidate parameters:

$$\sup_{\theta \in \Theta} |\widehat{Q}_M(\theta) - Q_M(\theta)| \to_p 0.$$

Estimation proof: discontinuity at zero information cost

Fix (α, δ, G, w) at a market with $\pi(z, x) = 0$, and let the candidate cost satisfy $\lambda_M(x) > 0$ and $\lambda_M(x) \to 0$.

- At every positive cost, the zero input share gives
$$\Psi\big((\alpha, \delta, \lambda_M, G); w, 0\big) = 0.$$
- At zero cost, the share map becomes
$$\Psi\big((\alpha, \delta, 0, G); w, 0\big) = \mathbb{P}_{\Omega \sim G}(\alpha + \delta z + \Omega > 0),$$
which can be positive. The squared residual can therefore jump.

Estimation proof: using an auxiliary criterion

- For the proof, keep only markets with interior population shares:

$$Q_M^\circ(\theta) = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\{0 < s_m < 1\}\, r(\theta; W_m, s_m)^2.$$

  This auxiliary criterion satisfies $0 \leq Q_M^\circ(\theta) \leq Q_M(\theta)$.

- Its population counterpart, with $s(w) = \pi(z, x)$, is

$$Q^\circ(\theta) = \mathbb{E}\left[\mathbf{1}\{0 < s(W) < 1\} r(\theta; W, s(W))^2\right].$$

Estimation proof: continuity at interior shares

- Fix $q = \pi(z, x) \in (0, 1)$. As $\lambda_M(x) \downarrow 0$,

$$\Psi\big((\alpha, \delta, \lambda_M, G); w, q\big)$$
$$\longrightarrow \mathbb{P}_{\Omega \sim G}(\alpha + \delta z + \Omega > 0) + q\, \mathbb{P}_{\Omega \sim G}(\alpha + \delta z + \Omega = 0).$$

- For a fixed candidate, $\delta \neq 0$ and a continuous $Z \mid X$ make exact payoff ties occur with probability zero across markets.

- Under the maintained conditions, almost-sure continuity and boundedness of the auxiliary loss give continuity of Q°.

Estimation proof: the population loss vanishes

- Boundedness, almost-sure continuity at each candidate, and compactness give

$$\sup_{\theta \in \Theta} |Q_M^\circ(\theta) - Q^\circ(\theta)| \to_p 0.$$

- Approximate minimization, negligible share-substitution error, and a feasible sieve approximation give

$$0 \leq Q_M^\circ(\widehat{\theta}_M) \leq Q_M(\widehat{\theta}_M) \leq Q_M(\theta_M^*) + o_p(1) = o_p(1).$$

- Uniform convergence therefore implies

$$Q^\circ(\widehat{\theta}_M) \to_p 0.$$

Estimation proof: separating the true parameter

- Identification makes θ_0 the unique zero:

$$Q^\circ(\theta) = 0 \quad \Longleftrightarrow \quad \theta = \theta_0.$$

- Continuity and compactness separate every other parameter from that zero:

$$\inf_{d(\theta,\theta_0)\geq\varepsilon} Q^\circ(\theta) > 0 \qquad (\varepsilon > 0).$$

- Since $Q^\circ(\widehat{\theta}_M) \to_p 0$, the estimator must approach the truth:

$$d(\widehat{\theta}_M, \theta_0) \to_p 0.$$

Inference: binding constraints

- **Binding parameter constraints can make the estimator's limiting distribution nonnormal.**
  - A sieve approximates the taste-shock distribution using nonnegative probability weights on a grid or histogram bins. Some weights may equal zero.
  - Information costs cannot be negative and may equal zero.
- **I use test inversion for inference.**
  I collect parameter values that a test based on the RI residuals does not reject. Taking the range of a scalar function over this region gives an interval for a parameter or a counterfactual.

Inference: test statistic

$$T_M(\theta) := \sum_{m \in \widehat{\mathcal{A}}_M(\theta)} \frac{N_M \, r_m(\theta, \widehat{s}_m)^2}{A_m(\theta, \widehat{s}_m)^2 \widehat{s}_m (1 - \widehat{s}_m)}.$$

- Each market has N_M choices; $\widehat{s}_m$ is its observed share. Define
$$r_m(\theta, q) := r(\theta; W_m, q), \qquad A_m(\theta, q) := \partial_q r_m(\theta, q).$$
- For fixed $\eta \in (0, 1/2)$ and $a_* > 0$, retain the markets
$$\widehat{\mathcal{A}}_M(\theta) := \{m : \eta \le \widehat{s}_m \le 1 - \eta, \ |A_m(\theta, \widehat{s}_m)| \ge a_*\}.$$
- Trimming prevents shares near zero or one and small residual derivatives from magnifying individual terms in $T_M(\theta)$.

Inference: confidence regions and intervals

- For $\gamma \in (0,1)$, let $\chi^2_{1-\gamma,k}$ be the $(1-\gamma)$ chi-square quantile with k degrees of freedom.
- Collect sieve parameters that the test does not reject:

$$\mathcal{C}^s_{M,\gamma} := \left\{ \theta \in \Theta_M : T_M(\theta) \leq \chi^2_{1-\gamma,|\widehat{\mathcal{A}}_M(\theta)|} \right\}.$$

- For a scalar function $\phi : \Theta \to \mathbb{R}$, report $[L_M, U_M]$, where

$$L_M := \inf_{\theta \in \mathcal{C}^s_{M,\gamma}} \phi(\theta), \qquad U_M := \sup_{\theta \in \mathcal{C}^s_{M,\gamma}} \phi(\theta).$$

  Examples are $\phi(\theta) = \delta$, $\phi(\theta) = \lambda(x)$ at a fixed x, and counterfactual changes in choice probabilities.
- Validity does not require point identification. Projection gives conservative coverage under the conditions that follow.

Formal conditions Continue to the application

Inference: sampling and trimming assumptions

- **Assumption F1: Joint growth**
  Each market has N_M choices, with

$$M, N_M \to \infty, \qquad M/\sqrt{N_M} \to 0.$$

- Fix $\eta \in (0, 1/2)$ and $a_* > 0$. Write

$$s(w) := \pi(z, x), \qquad A_0(w) := \partial_q r(\theta_0; w, s(w)),$$

  with the derivative defined for $0 < s(w) < 1$.

- **Assumption F2: Positive probability of satisfying the trimming conditions**

$$\mathbb{P}\{\eta < s(W) < 1 - \eta,\ |A_0(W)| > a_*\} > 0.$$

Inference: trimming with population probabilities

- Take a deterministic $\theta_M^* \in \Theta_M \subseteq \Theta$ with $d(\theta_M^*, \theta_0) \to 0$, as in E3.
- For $0 < s(w) < 1$, let $A_M(w) := \partial_q r(\theta_M^*; w, s(w))$. Define the set using population probabilities:

$$\mathcal{R}_M := \{w : \eta \leq s(w) \leq 1 - \eta,\ |A_M(w)| \geq a_*\}.$$

- For $0 < s(w) < 1$, define

$$\Delta_M(w) := \min\left\{|s(w) - \eta|,\ |s(w) - (1-\eta)|,\ \left||A_M(w)| - a_*\right|\right\}.$$

  Set $\Delta_M(w) := \eta$ for $s(w) \in \{0, 1\}$.
- **Assumption F3: Little probability near the trimming cutoffs**
  There exist $C < \infty$, $t_0 > 0$ and an integer M_0 such that

$$\mathbb{P}\{\Delta_M(W) \leq t\} \leq Ct \quad \text{for every } M \geq M_0, \quad 0 < t \leq t_0.$$

Inference: sieve approximation assumptions

- **Assumption F4: Negligible residual bias**
  Define
  $$\rho_M := \sup_{w \in \mathcal{R}_M} |r(\theta_M^*; w, s(w))|.$$
  Require $M^{1/4}\sqrt{N_M}\,\rho_M \to 0$.

- **Assumption F5: Negligible approximation error in the reported function**
  Let $\phi_M^* := \phi(\theta_M^*)$ and $E_M := \{\theta_M^* \in \mathcal{C}_{M,\gamma}^s\}$. For some deterministic $\varepsilon_M \downarrow 0$,
  $$|\phi_M^* - \phi(\theta_0)| = o(\varepsilon_M),$$
  $$\mathbb{P}\{[\phi_M^* - \varepsilon_M, \phi_M^* + \varepsilon_M] \subseteq [L_M, U_M] \mid E_M\} \to 1.$$

Inference: asymptotic validity

Theorem 3 Under the stated sampling assumptions, E3, ID1(b), ID3(a), and F1–F5, with compact Θ and continuous $\phi : \Theta \to \mathbb{R}$,

$$\liminf_{M \to \infty} \mathbb{P}\{\phi(\theta_0) \in [L_M, U_M]\} \geq 1 - \gamma.$$

Empirical Application: Amazon Mechanical Turk

- I use my method to study how information costs affect task-taking probabilities in online labour markets.
- Amazon Mechanical Turk provides an application using the data of Dubé et al. (2020).
- Workers see a posted wage and decide whether to take the task. Assessing its difficulty and personal suitability requires cognitive effort.
- I estimate how assessment costs vary with task descriptions, then compare wage responses under different information costs.

Empirical Application: task descriptions

- **Task specificity:** a concrete action and its target.
- **Workload specificity:** a stated duration, numerical amount of work, or effort limit.

x	Category	Illustrative description
0	Both specific	Find the agent's email address; inspect up to three search results.
1	Task only	Find the agent's email address.
2	Workload only	Complete a two-minute study.
3	Neither specific	Complete a survey.

Measuring task specificity

Empirical Application: interpretation

- **What is the signal?**
  An internal assessment of the task's difficulty and personal suitability, formed by processing the posting and drawing on experience.
- **Why might specificity reduce λ?**
  - A concrete task tells the worker what to imagine doing.
  - A specified workload gives that mental simulation an extent or stopping point.
- A given improvement in understanding the payoff can then require less cognitive effort.

Empirical Application: specification

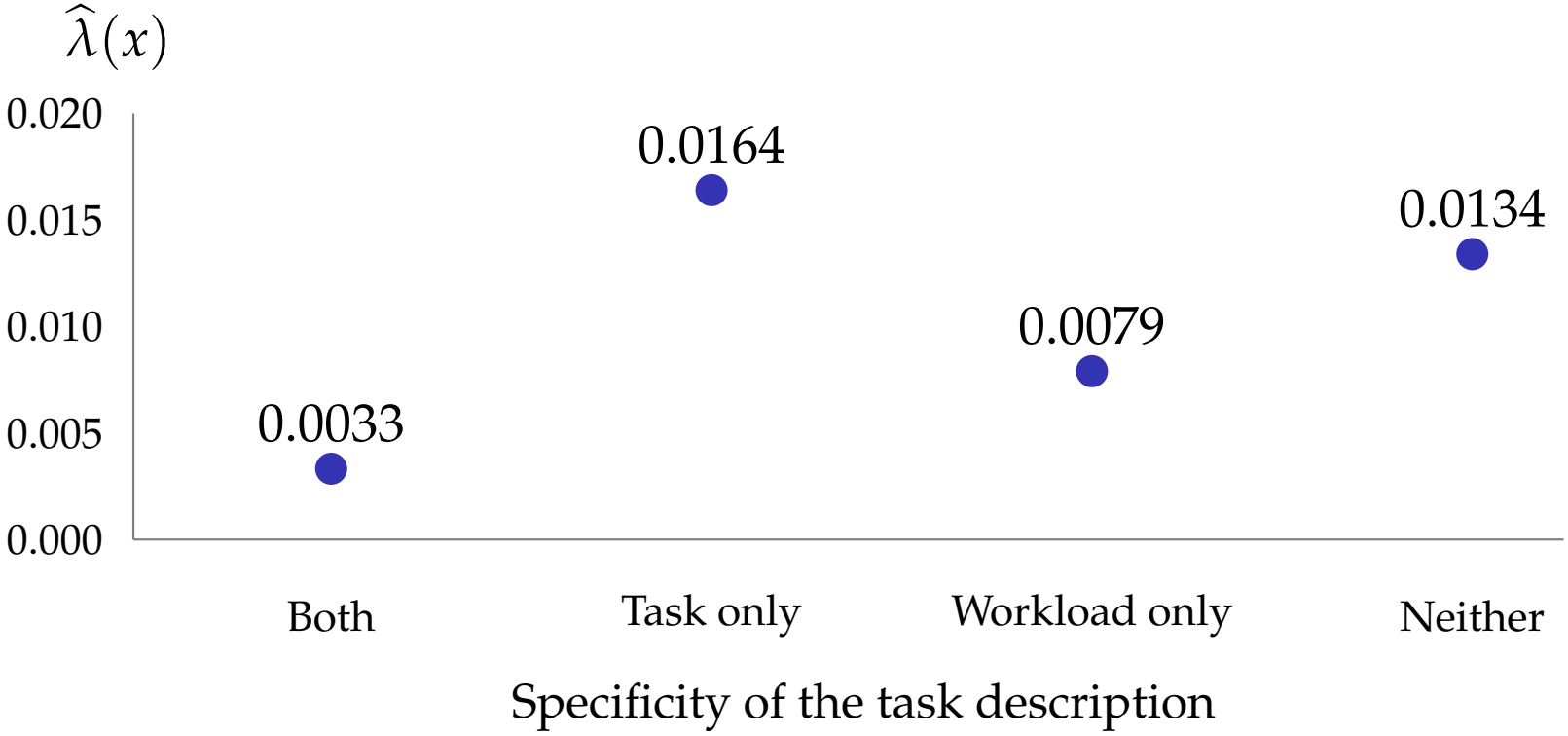
Worker i chooses between taking task m, with payoff v_{im}, and not taking it, with payoff normalized to zero.

$$\boxed{v_{im} = a_{c(m)} + \delta z_m + \Omega_{im}, \qquad \lambda_m = \lambda(x_m).}$$

- $c(m)$ groups posting m with postings from the same requester that have matching task descriptions.
- $a_{c(m)}$: average attractiveness of that task; z_m: the reward index.

Sample construction

Empirical Application: estimated information costs



Empirical Application: CIs for the estimated costs

Quantity	Estimate	Actual CI	Calibrated CI
Both specific	0.0033	$[0,\ 8]$	$[0,\ 0.0555]$
Task only	0.0164	$[0,\ 0.5953]$	$[0.0123,\ 0.3017]$
Workload only	0.0079	$[0,\ 2.1085]$	$[0.0031,\ 0.1107]$
Neither specific	0.0134	$[0,\ 8]$	$[0.0043,\ 0.2540]$
Wage coefficient δ	0.0017	$[10^{-6},\ 1.2694]$	$[0.0015,\ 0.0421]$

Numerical approximations to 95% Poisson Monte Carlo test-inversion intervals.
Calibrated data: 1,370 simulated postings and 100 times exposure.

Empirical Application: doubling wages

Task-taking probabilities when wages double for all 137 postings.

Information regime	Before (%)	After (%)	Change (pp)	Elasticity
Estimated costs	2.62	3.20	0.577	0.287
Half costs	8.90	9.32	0.426	0.068
Full information	12.87	13.21	0.335	0.037

Counterfactual CIs

Conclusion

- **Identification**
  Choice-curve moments identify utility coefficients and information costs. Deconvolution identifies the taste distribution.
- **Estimation**
  An identifying auxiliary criterion handles discontinuity and establishes sieve consistency from aggregate shares.
- **Inference**
  Test inversion gives confidence intervals for structural counterfactuals under conditions controlling approximation error.
- Thank you.

Appendix: sample construction

The postings were observed between June 1 and August 20, 2017, in the data collected by Dubé et al. (2020).

- **Usable records:** positive wage, identified requester, at least two inventory observations in the first 30 minutes, and sufficient calibrated exposure.
- **Initial inventory:** at least 10 assignments.
- **Repeated tasks with wage variation:** same requester, matching task descriptions and qualifications; at least four postings, three distinct wages, and a highest wage at least twice the lowest. This leaves **174 postings**.
- **Common observation window:** require inventory observations through exactly three hours. The final sample has **137 postings in 35 requester–task groups**.

Appendix: measuring task specificity

- Review each posting's title and description together: **44 distinct texts** in the sample.
- **Task:** a concrete action and target. "Find an email address" qualifies; a generic "survey" or study topic does not.
- **Workload:** a numerical duration, quantity of work, or explicit effort limit. "Two minutes" qualifies; "quick" alone does not.
- Cross the two indicators: both, task only, workload only, or neither specific. The categories contain **29, 83, 9, and 16 postings**.
- A saved coding table assigns identical text identically. It is a proxy for assessment difficulty, not a direct measure of cognitive cost.

Appendix: task taking and Poisson thinning

- Potential arrivals follow $N_m \sim \text{Poisson}(E_m)$. If each worker independently takes the task with probability b_m,

$$A_m \sim \text{Poisson}(E_m b_m), \qquad \widehat{q}_m = A_m / E_m.$$

$$\text{Var}(\widehat{q}_m \mid W_m) = b_m / E_m.$$

- Proxy tasks taken by positive inventory decreases over three hours. Exposure is calibrated as $E_m = 20B_{m0}$, with initial inventory B_{m0}; arrivals are not observed.
- Both datasets use $\eta = .002$ and $D_0 = 10^{-5}$. Calibrated data: 1,370 independent count observations, with $100E_m$ per posting and the same 35 task groups.
- The calibrated CIs concern a hypothetical larger dataset generated at the fitted parameters.

Appendix: CIs for counterfactuals

Quantity	Estimate	Actual CI*	Calibrated CI*
Wage effect (pp)	0.577	[0.000, 18.675]	[0.527, 1.299]
Full-information effect (pp)	10.253	[−4.390, 14.945]	[3.970, 14.532]
Estimated-cost elasticity	0.287	[0.000, 2.353]	[0.072, 0.305]
Half-cost elasticity	0.068	[0.000, 2.196]	[0.025, 0.196]
Full-information elasticity	0.037	[0.000, 1.931]	[0.010, 0.131]
Elasticity difference	0.250	[−0.504, 1.627]	[0.000, 0.277]

*Numerical approximations to 95% Poisson Monte Carlo test-inversion intervals; global endpoints are not certified.

Full-information effect: zero versus estimated costs at original wages.
Elasticity difference: estimated costs minus full information.

Appendix: computing the confidence intervals

- For each full candidate parameter, compute the Poisson-standardized RI residual statistic.
- Simulate 3,999 datasets under that candidate, keeping design and exposure fixed. Recompute trimming in every simulation and use the simulated rank to test the candidate.
- Project the accepted parameter vectors onto each quantity using SLSQP from multiple starts. All nuisance parameters remain free; $0 \leq \lambda \leq 8$.

The application uses simulated critical values. The earlier theorem uses analytic critical values. Empty retained sets are accepted.

Monte Carlo test principle: Dufour (2006). Computational details and the ordering statistic are in the replication notes.

Appendix: numerical checks

- The full fitted model is accepted in the actual data: Monte Carlo $p \approx 0.177$.
- Every reported endpoint has a saved full parameter vector satisfying the model constraints and the primary test.
- Endpoint acceptance is unchanged in an independent batch of 3,999 simulations. Both datasets use the same test and trimming rule.

These checks support reproducibility. Numerical projections can miss accepted components; the calibrated column is a larger-sample illustration.